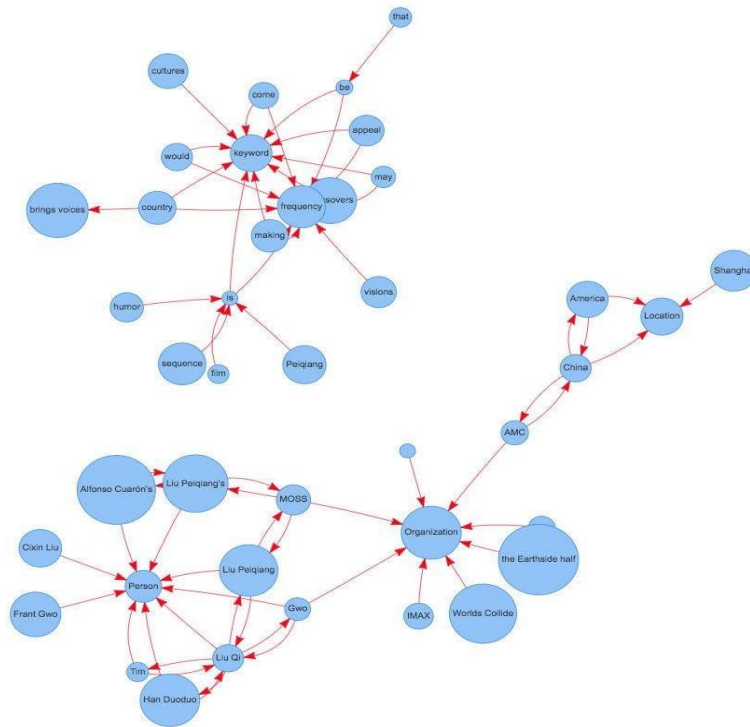


Aim: Representation of a document with their keywords using TextRank

IDE: Google Colab

Theory:

TextRank is an algorithm based on PageRank, which often used in keyword extraction and text summarization. In this article, I will help you understand how TextRank works with a keyword extraction example and show the implementation by Python.

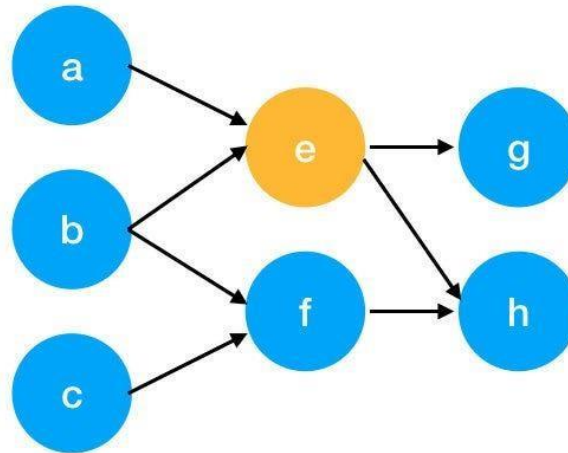


Understand PageRank

There are tons of articles talking about PageRank, so I just give a brief introduction to PageRank. This will help us understand TextRank later because it is based on PageRank. PageRank (PR) is an algorithm used to calculate the weight for web pages. We can take all web pages as a big directed graph. In this graph, a node is a webpage. If webpage A has the link to web page B, it can be represented as a directed edge from A to B. After we construct the whole graph, we can assign weights for web pages by the following formula.

$$S(V_i) = (1 - d) + d * \sum_{j \in In(v_i)} \frac{1}{|Out(V_j)|} S(V_j)$$

- $S(V_i)$ - the weight of webpage i
- d - damping factor, in case of no outgoing links
- $In(V_i)$ - inbound links of i, which is a set
- $Out(V_j)$ - outgoing links of j, which is a set
- $|Out(V_j)|$ - the number of outbound links



Here is an example to better understand the notation above. We have a graph to represent how web pages link to each other. Each node represents a webpage, and the arrows represent edges. We want to get the weight of webpage e. We can rewrite the summation part in the above function to a simpler version.

$$\begin{aligned}
 In(v_e) &= \{a, b\}, j \in \{a, b\} \\
 \sum_{j \in \{a, b\}} \frac{1}{|Out(V_j)|} S(V_j) &= \frac{1}{|Out(V_a)|} S(V_a) + \frac{1}{|Out(V_b)|} S(V_b) \\
 &= \frac{1}{|\{e\}|} S(V_a) + \frac{1}{|\{e, f\}|} S(V_b) \\
 &= S(V_a) + \frac{1}{2} S(V_b)
 \end{aligned}$$

We can get the weight of webpage e by the following function.

$$S(V_e) = (1 - d) + d * \left(S(V_a) + \frac{1}{2} S(V_b) \right)$$

We can see the weight of the webpage e is dependent on the weights of its inbound pages. We need to run this iteration much time to get the final weight. In the initialization, the importance of each webpage is 1.

Pre Lab Exercise:

1. What are the key components of TextRank?

- Text preprocessing
- Graph construction
- Co-occurrence window
- Edge weighting
- PageRank algorithm (ranking)
- Convergence
- Keyword selection

2. What are some limitations of TextRank for keyword extraction?

- Does not understand semantic meaning .
- May extract irrelevant or generic words.
- Performance depends on text preprocessing quality.
- Not effective for very short texts.
- Cannot capture deep context or meaning like modern NLP models.
- May produce duplicate or similar keywords.

3. What are the advantages of TextRank for keyword extraction?

- Unsupervised method
- Simple and easy to implement.
- Works well for keyword extraction and summarization.
- Domain-independent .
- Based on a strong algorithm.
- Efficient for medium to large documents.

Program (Code):

```
!pip install summa
from summa import keywords
text = ""
```

```
I am a B.Tech student in Information and Communication Technology at Marwadi
University.
```

```
I am interested in Artificial Intelligence, Machine Learning, and Natural Language
Processing.
```



Marwadi University
Faculty of Technology
Department of Information and Communication Technology

Subject: Artificial Intelligence (01CT0616)

Aim: Representation of a document with their keywords using TextRank

Experiment No: 09

Date:19/03/2026

Enrolment No:92301733054

Currently, I am working on an AI-based answer sheet evaluation system using OCR and NLP.

I have knowledge of Python, Flutter, and Laravel development.

I enjoy building real-world applications in education and healthcare.

I also like photography, content creation, and social media marketing.

"""

```
kw = keywords.keywords(text, words=25)
```

```
print("Top Keywords:\n")
```

```
print(kw)
```

```
kw_list = keywords.keywords(text, words=25, split=True)
```

```
print("Top 25 Keywords List:\n")
```

```
print(kw_list)
```

```
for i, word in enumerate(kw_list, 1):
```

```
    print(f"{i}. {word}")
```

Results:

```
for i, word in enumerate(kw_list, 1):  
    print(f"{i}. {word}")
```

```
... 1. b student  
    2. intelligence  
    3. machine  
    4. enjoy building  
    5. real applications  
    6. learning  
    7. artificial  
    8. social media  
    9. photography content  
    10. marketing  
    11. technology  
    12. communication  
    13. python  
    14. flutter  
    15. university  
    16. marwadi  
    17. like  
    18. creation
```

Observation: I have observed that the TextRank algorithm effectively extracts important keywords from the given text based on word relationships and frequency. The extracted keywords represent the main topics such as AI, NLP, and development, but some generic or less meaningful words are also included. Overall, the method provides a good summary of the document, though it does not fully capture the contextual meaning.

Post Lab Exercise:

1. Write about “your ownself” (in not more than 500 words You know better about you, rather than ChatGPT!!). Extract top-25 keywords for your portfolio and analyze it in 3 categories: keyword can not give the real idea, keyword gives the real idea, keyword is irrelevant. Paste the code, your portfolio, output and your analysis.



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Experiment No: 09

Date:19/03/2026

Enrolment No:92301733054

Programme Code:

```
!pip install summa
from summa import keywords
text = """
I am a B.Tech student in Information and Communication Technology at Marwadi University.
I have a strong interest in Artificial Intelligence, Machine Learning, and Natural Language Processing.
Currently, I am working on an AI-assisted answer sheet evaluation system that uses OCR and NLP techniques to analyze student answers.
I have knowledge of programming languages such as Python, Dart, and PHP, and I have experience working with frameworks like Flutter and Laravel.
I enjoy developing real-world applications, especially in the education and healthcare domains.
Apart from academics, I am also interested in photography and content creation. I like capturing natural scenes, especially sunsets, and creating aesthetic social media content.
I also work with saree-related content and marketing.
I am a hardworking and curious learner who enjoys exploring new technologies and solving practical problems.
My goal is to build innovative AI-based solutions that can make real-world impact.
"""

kw = keywords.keywords(text, words=25)

print("Top Keywords:\n")
print(kw)
kw_list = keywords.keywords(text, words=25, split=True)

print("Top 25 Keywords List:\n")
print(kw_list)
for i, word in enumerate(kw_list, 1):
    print(f"{i}. {word}")
```

Output:

```
for i, word in enumerate(kw_list, 1):
    print(f"{i}. {word}")

1. content
2. like
3. student
4. language
5. languages
6. answers
7. real
8. practical
9. domains
10. especially
11. natural
12. ai answer sheet
13. machine
14. intelligence
15. technology
16. new technologies
17. aesthetic social media
18. innovative
19. processing
```